

BML lecture #7: Generative priors

<http://github.com/rbardenet/bml-course>

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Generative priors: Using generative models to solve (Bayesian) inverse problems.

- ▶ Part I: A fast introduction to score based generative model (SGM)
- ▶ Part II: How to use SGM as prior for inverse problems

The original miracle!



Figure: $X_0 \sim p_{\mathcal{D}}$.



Figure: $X_{\sigma} = X_0 + \sigma Z$.¹

- ▶ D_{θ_0} untrained CNN,
- ▶ $L_{\sigma}(\theta) = \|D_{\theta}(X_{\sigma}) - X_{\sigma}\|^2$, $L_0(\theta) = \|D_{\theta}(X_{\sigma}) - X_0\|^2$,
- ▶ $\theta_t = \theta_{t-1} - \nabla L_{\sigma}(\theta_{t-1})$

What happens to $\{L_{\sigma}(\theta_t)\}_{t=1}^N$ and $\{L_0(\theta_t)\}_{t=1}^N$?

¹ $Z \sim \mathcal{N}(0, I)$, $X_0 \perp Z$.

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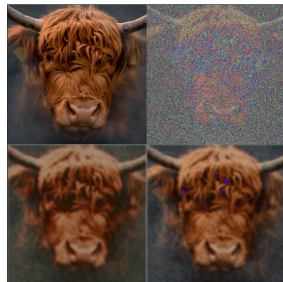
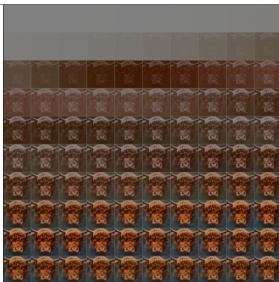
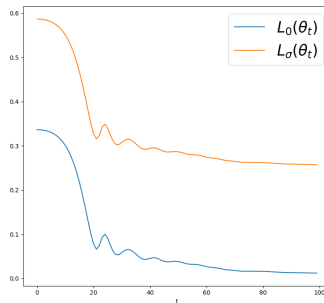
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The original miracle ².



²This experiment is inspired by Ulyanov, Vedaldi, and Lempitsky, 2018

How to can we take advantage of extremely good denoisers to create a generative model?

Lemma (Vincent, 2011)

→ Optimal denoiser! (Definition of cond. exp. for L^2 variables)

$$\mathbb{E}[X_0 | X_\sigma = x] = x + \sigma^2 \nabla \log p_\sigma(x), \quad (1)$$

where $p_\sigma = \text{Law}(X_\sigma)$.

$$p_\sigma(x) = \int \underbrace{N(x; x_0, \sigma^2 I_d)}_{:= p_\sigma(x|x_0)} p_0(x_0) dx_0.$$

$$\nabla \log p_\sigma(x) = \frac{\nabla p_\sigma(x)}{p_\sigma(x)} = \frac{\int \nabla p_\sigma(x|x_0) p_0(x_0) dx_0}{\frac{p_\sigma(x)}{p_\sigma(x)}} = \int \nabla \log p_\sigma(x|x_0) \underbrace{\frac{p_\sigma(x|x_0) p_0(x_0)}{p_\sigma(x)}}_{\text{Density of } X_0 | X_\sigma} dx_0$$

$$\Rightarrow \nabla \log p_\sigma(x) = \mathbb{E}[\nabla \log p_\sigma(x|x_0) | X_\sigma = x]. \quad \nabla \log p_\sigma(x|x_0) = -\frac{(x - x_0)}{\sigma^2}$$

$$\Rightarrow \nabla \log p_\sigma(x) = \frac{-(x - \mathbb{E}[X_0 | X_\sigma = x])}{\sigma^2}$$

$$0 < \sigma_1 < \cdots < \sigma_T.$$

$$\theta \in \arg \min \sum_{t=1}^N \gamma_t^2 \mathbb{E} [\|D_\theta(X_0 + \sigma_t Z, \sigma_t) - X_0\|^2] \quad (2)$$

Sequential Langevin Algorithm:

$X_T \sim \mathcal{N}(0, \sigma_T^2 \mathbf{I})$, define $s_\theta(x, \sigma) = \sigma^{-2}(D_\theta(x, \sigma) - x)$.

For $t = T - 1, \dots, 1$

- ▶ $X_{0,t} = X_{K,t+1}$
- ▶ $X_{k,t} = X_{k-1,t} + \eta s_\theta(X_{k-1,t}, \sigma) + \sqrt{2\eta} Z_{k,t}.$

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Problem: Neural function evaluation (NFE) = TK

Consider the following Markov chain:

- 1 $X_0 \sim p_{\mathcal{D}},$

- 2 $X_t \sim X_{t-1} + \sqrt{\sigma_t^2 - \sigma_{t-1}^2} \cdot Z_t$

The joint law of this chain is

$$p_{0:T}(x_{0:T}) = p_{\mathcal{D}}(x_0) \prod_{t=1}^T \overbrace{\mathcal{N}(x_t; x_{t-1}, (\sigma_t^2 - \sigma_{t-1}^2)\mathbf{I})}^{:= p_{t|t-1}(x_t|x_{t-1})}. \quad (3)$$

It holds that:

- 1 $p_{t|0}(x_t|x_0) = \mathcal{N}(x_0, \sigma_t^2\mathbf{I}),$

- 2 $p_{t|0,t+1}(x_t|x_0, x_{t+1}) = \mathcal{N}(x_t, x_0 + \frac{\sigma_t}{\sigma_{t+1}^2}(x_{t+1} - x_0), \frac{\sigma_t^2}{\sigma_{t+1}^2}(\sigma_{t+1}^2 - \sigma_t^2)\mathbf{I})$

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The joint law of this chain is

$$p_{0:T}(x_{0:T}) = p_D(x_0) \prod_{t=1}^T \overbrace{\mathcal{N}(x_t; x_{t-1}, (\sigma_t^2 - \sigma_{t-1}^2)I)}^{:= p_{t|t-1}(x_t|x_{t-1})}. \quad (3)$$

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$$\underbrace{\text{Law}(X_t | X_0, X_{t+1})}_{\text{Law}(X_t | X_0, X_{t+1})} = \frac{p_{t+1|t}(x_{t+1}|x_t) p_{t|0}(x_t|x_0)}{p_{t+1|0}(x_{t+1}|x_0)}$$

Gaussians that conjugate well!

We want to reverse the noising process. Consider

$$q_{0:T}^{\theta}(x_{0:T}) := \mathcal{N}(x_T; \mu_T, \eta_T^2 \mathbf{I}) \prod_{t=0}^{T-1} \mathcal{N}(x_t; \mu_{t,\theta}(x_{t+1}), \eta_{t+1}^2 \mathbf{I}) . \quad (4)$$

Our goal is to

$$\arg \min_{\theta, \eta_{0:T}^2, \mu_T} D_{\text{KL}}(p_{0:T} || q_{0:T}^{\theta}) . \quad (5)$$

Before advancing, note that $p_{0:T}$ (as it is a Markov chain) admits the following decomposition

$$p_{0:T}(x_{0:T}) = p_T(x_T) \prod_{t=0}^{T-1} p_{t|t+1}(x_t | x_{t+1}) .$$

Unfortunately, each $p_{t|t+1}$ is intractable for general p_D .

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Unfortunately, each $p_{t|t+1}$ is intractable for general $p_{\mathcal{D}}$.

$$\begin{aligned}
 D_{\text{KL}}(p_{0:T} || q_{0:T}^\theta) &= \\
 \mathbb{E}_{p_{0:T}} \left[\log \left(\frac{p_T(x_T) \prod_{t=0}^{T-1} p_{t|t+1}(x_t | x_{t+1})}{\mathcal{N}(x_T; \mu_T, \eta_T^2 \mathbf{I}) \prod_{t=0}^{T-1} \mathcal{N}(x_t; \mu_{t,\theta}(x_{t+1}), \eta_{t+1}^2 \mathbf{I})} \right) \right] \\
 &= D_{\text{KL}}(p_T || \mathcal{N}(\mu_T, \eta_T^2 \mathbf{I})) \\
 &\quad + \sum_{t=1}^{T-1} \mathbb{E}_{p_{0:T}} [D_{\text{KL}}(p_{t|t+1}(\cdot | X_{t+1}) || \mathcal{N}(\mu_{t,\theta}(X_{t+1}), \eta_{t+1}^2 \mathbf{I}))] \\
 &\quad + \mathbb{E}_{p_{0:T}} [D_{\text{KL}}(p_{\mathcal{D}} || \mathcal{N}(\mu_{0,\theta}(X_1), \eta_{t+1}^2 \mathbf{I}))] .
 \end{aligned}$$

We now use the following property:

Lemma

If $\lambda \in \mathcal{P}_2$, then

$$(\mu_\lambda, \Sigma_\lambda) \in \arg \min D_{\text{KL}}(\lambda || \mathcal{N}(\mu, \Sigma)) ,$$

where $\mu_\lambda = \mathbb{E}_\lambda[X]$ and $\Sigma = \mathbb{V}_\lambda[X] = \mathbb{E}_\lambda[(X - \mu_\lambda)(X - \mu_\lambda)^T]$.

$$\begin{aligned}
 D_{\text{KL}}(p_{0:T} || q_{0:T}^\theta) &= \\
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By the previous lemma, we just need to calculate $\mathbb{E}[X_T], \mathbb{V}[X_T]$.

$$1. \mathbb{E}[X_T] = \mathbb{E}[\mathbb{E}[X_T | X_0]] = \mathbb{E}[X_0] = 0 \quad \rightarrow \text{we always assume this.}$$

$$2. \mathbb{V}[X_T] = \mathbb{V}[\mathbb{E}[X_T | X_0]] + \mathbb{E}[\mathbb{V}[X_T | X_0]] = \mathbb{V}[X_0] + \sigma_T^2 = 1 + \sigma_T^2$$

$$\Rightarrow \mu_T = 0, \eta_T^2 = \sigma_T^2 + 1$$

$$\mathbb{E}_{p_{0:T}} [\text{D}_{\text{KL}} (p_{t|t+1}(\cdot|X_{t+1}) || \mathcal{N}(\mu_{t,\theta}(X_{t+1}), \eta_{t+1}^2 \mathbf{I}))]$$

Again,

$$\begin{aligned} 1. \mathbb{E}[X_t | X_{t+1}] &= \mathbb{E}[\mathbb{E}[X_t | X_{t+1}, X_0] | X_{t+1}] = \mathbb{E}\left[\left(1 - \frac{\sigma_t^2}{\sigma_{t+1}^2}\right) X_0 + \frac{\sigma_t^2}{\sigma_{t+1}^2} X_{t+1} \mid X_{t+1}\right] \\ &= \left(1 - \frac{\sigma_t^2}{\sigma_{t+1}^2}\right) \underbrace{\mathbb{E}[X_0 | X_{t+1}]}_{\substack{\uparrow \\ \text{Demoisier}}} + X_{t+1} \frac{\sigma_t^2}{\sigma_{t+1}^2} \Rightarrow \mu_{t,\theta}(x) = \left(1 - \frac{\sigma_t^2}{\sigma_{t+1}^2}\right) D_\theta(x, \sigma_{t+1}) + \frac{\sigma_t^2}{\sigma_{t+1}^2} x \end{aligned}$$

$$\begin{aligned} 2. \mathbb{V}[X_t | X_{t+1}] &= \mathbb{V}[\mathbb{E}[X_t | X_{t+1}, X_0] | X_{t+1}] + \mathbb{E}[\mathbb{V}[X_t | X_{t+1}, X_0] | X_{t+1}] \\ &= \mathbb{V}\left[\left(1 - \frac{\sigma_t^2}{\sigma_{t+1}^2}\right) X_0 + \frac{\sigma_t^2}{\sigma_{t+1}^2} X_{t+1} \mid X_{t+1}\right] + \frac{\sigma_t^4}{\sigma_{t+1}^2} (\sigma_{t+1}^2 - \sigma_t^2) \\ &\Rightarrow \underbrace{\left(1 - \frac{\sigma_t^2}{\sigma_{t+1}^2}\right)^2 \mathbb{V}[X_0 | X_{t+1}]}_{\substack{\text{Hard to} \\ \text{approximate!}}} + \frac{\sigma_t^2}{\sigma_{t+1}^2} (\sigma_{t+1}^2 - \sigma_t^2) \end{aligned}$$

$$\eta_t^2 = \frac{\sigma_t^2}{\sigma_{t+1}^2} (\sigma_{t+1}^2 - \sigma_t^2)$$

$$\mathbb{E}_{p_{0:T}} [D_{KL}(p_D || \mathcal{N}(\mu_{0,\theta}(X_1), \eta_0^2))]$$

$$\begin{aligned} \mathbb{E}[D_{KL}(p_D || \mathcal{N}(\mu_{0,\theta}(X_1), \eta_0^2))] &= \mathbb{E}_{p_D}[\log p_D] - \mathbb{E}[\log \mathcal{N}(X_0, \mu_{0,\theta}(X_1), \eta_0^2)] \\ &= \mathbb{E}_{p_D} \left[\frac{\|X_0 - \mu_{0,\theta}(X_1)\|^2}{2\eta_0^2} \right] - \frac{d}{2} \log \eta_0^2 + \text{cte} \end{aligned}$$

We typically set $\mu_{0,\theta}(X_1) = D_\theta(X_1, \sigma_1)$ and $\eta_0^2 < \sigma_1$.

Algorithm:

1 Sample $X_T \sim \mathcal{N}(0, \sigma_T^2)$,

2 Sample for $t = T - 1, \dots, 1$,

$$X_t \sim \mathcal{N} \left(D_\theta(X_{t+1}, \sigma_{t+1}) + \frac{\sigma_t^2}{\sigma_{t+1}^2} (X_{t+1} - D_\theta(X_{t+1}, \sigma_{t+1})), \frac{\sigma_t^2}{\sigma_{t+1}^2} (\sigma_{t+1}^2 - \sigma_t^2) \mathbf{I} \right) .$$

Often in the literature, you see the following noising process for $\beta_t \in (0, 1)$:

$$Y_t = \sqrt{1 - \beta_t} Y_{t-1} + \sqrt{\beta_t} Z_t . \quad (6)$$

It has the nice property that $\mathbb{E}[Y_t] = 0$ and $\mathbb{V}[Y_t] = 1$ (Thus variance preserving) and that

$$p(y_t|y_0) = \mathcal{N}(\sqrt{\alpha_t} y_0, (1 - \alpha_t) \mathbf{I})$$

where $\alpha_t = \prod_{k \leq t} (1 - \beta_k)$.

It can be obtained from our process by defining $Y_t = X_t / \sqrt{\sigma_t^2 + 1}$.

Furthermore, knowing the score of X_t can give access to the score of Y_t !
(The two are equivalent).

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Furthermore, knowing the score of X_t can give access to the score of Y_t ! (The two are equivalent).

DDPM notation:

$$\alpha_t = 1 - \beta_t$$

$$\bar{\alpha}_t = \prod_{i=1}^t \alpha_i = \prod_{i=1}^t (1 - \beta_i)$$

Backward Sampling (Variance preserving)

Algorithm:

- 1 Sample $X_T \sim \mathcal{N}(0, I)$,
- 2 Sample for $t = T - 1, \dots, 1$,

$$X_t \sim \mathcal{N} \left(\frac{1}{\sqrt{1 - \beta_{t+1}}} (X_{t+1} - \frac{\beta_{t+1}}{\sqrt{1 - \alpha_t}} \varepsilon_{\theta}(X_{t+1}, \sqrt{\alpha_{t+1}})), \frac{1 - \alpha_t}{1 - \alpha_{t+1}} \beta_{t+1} I \right),$$

where the **epsilon net** ε_{θ} can be linked to our denoising network by:

$$D_{\theta}(x, \sigma(\alpha_t)) = \frac{1}{\sqrt{\beta_t}} \left(x - \frac{\beta_t}{\sqrt{1 - \alpha_t}} \varepsilon_{\theta}(X_{t+1}, \sqrt{\alpha_t}) \right),$$

where $\sigma(\alpha_t) = \sqrt{\frac{1 - \alpha_t}{\alpha_t}}$.

Remember our inverse problem posterior:

$$p_0(x|y) = \frac{\ell(y|x)p_0(x)}{\int \ell(y|\tilde{x})p_0(\tilde{x})d\tilde{x}} . \quad (7)$$

Can we use a diffusion model to sample from $p_0(x|y)$?

Remember our inverse problem posterior:

$$p_0(x|y) = \frac{\ell(y|x)p_0(x)}{\underbrace{\int \ell(y|\tilde{x})p_0(\tilde{x})d\tilde{x}}_{Z_y}}. \quad (7)$$

Can we use a diffusion model to sample from $p_0(x|y)$?

Without training!!

We start by defining

$$p_t(x|y) = \int p_0(x_0|y) \mathcal{N}(x; x_0, \sigma_t^2 I) dx_0. \quad (8)$$

↗ Backward kernel

What is the score?

$$p_t(x|y) = \int \frac{p_0(x_0) \ell(y|x_0)}{Z_y} p_{t|0}(x|x_0) dx_0 = \frac{p_t(x)}{Z_y} \underbrace{\int \ell(y|x_0) p_{t|0}(x_0|x) dx_0}_{\mathbb{E}[\ell(y|x_0) | X_t=x]}$$

$$\Rightarrow \nabla \log p_t(x|y) = \nabla \log p_t(x) + \nabla \log \mathbb{E}[\ell(y|x_0) | X_t=x]$$

Chung et al., 2023 proposes the following approximation

$$\nabla \log p_t(x|y) = \nabla \log p_t(x) + \zeta_t \nabla \log l(y| \underbrace{\mathbb{E}[X_0|X_t = x_t]}_{x + \sigma^2 \nabla \log p_t(x)}) \quad (9)$$

In practice, for linear problems (i.e.) $\log l(y|x) = -\frac{1}{2\sigma_y^2} \|y - Ax\|^2 + c$, the authors suggests setting $\zeta_t = \tilde{\zeta} \frac{2\sigma_y^2}{\|y - Ax\|}$ where $\tilde{\zeta}$ is a tunable constant (typically 1 or 0.1).

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